HOSPITAL RESOURCE OPTIMIZATION AND DEMAND FORECASTING

Project Id: 25-26J-226

Individual Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology
(Specializing in Information Technology)

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

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DECLARATION

I declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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(Dr. Nuwan Kodagoda)

ABSTRACT

In this significance study, a machine learning–based system is developed for medicine demand forecasting and accident surge detection to help organize hospital resources better. The management of medical resources is a major challenge due to the increasing patient demand, unexpected emergencies, and limited availability of supply.

Proposed system consists of two key modules: Medicine Demand Forecasting using Random Forest model and Accident surge detection using XGBoost model by analyzing historical medicine usage data with related records of accident-associated admissions, patterns are identified for predicting future demand. The system can detect spikes in accidents and automatically adapt the need for medicines, encouraging proactive planning to enhance emergency preparedness.

The system was designed in a web-based architecture (React as Front-end, Node. This includes js for the backend, and MongoDB as data storage. The evaluation results show that the forecasting is precise, as the average accuracy of our model was 95.45% and strong performance on surges of accidents.

This study brings to the forefront the need for predictive analytics adoption in healthcare systems to curb medicine shortages, wastage and operational efficiency. The proposed solution offers a scalable and data-driven method to optimize the management of hospital resources.

ACKNOWLEDGMENT

Most importantly I would like to convey my sincere gratitude for the help of educational supervisor Mr. Kanishka Yapa, who had guided me with support through all phases related to developing this undergraduate research from start until finish. Without his insightful feedback and expertise this study couldn´t have been developed to the point it is today.

I am also especially grateful to Mr. Uditha Dharmakeerthi, which was the co-supervisor of this research project for all his support and enthusiasms so that he keeps ready to provide assistance every time when it is required. That keeps this work until his measured feedback and constructive criticism allowed these challenges to be overcome.
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We thank Dr. Dissanayake for assistance provided during the data collection phase of this study, without which it would not have been possible to complete successfully this project. I also thank the staff of Yaggahapitiya Divisional Hospital for their cordiality and help in obtaining required data, during this research.

I also want to give a big shout out to my fellow research teammates for their teamwork, the hard work and support they put through the course of this project. The collaboration and dedication of their people helped us achieve our common goals.

Last but not least, my thanks to my family and friends for putting up with me throughout this strenuous yet fulfilling experience. My sincere thanks to all who had a direct or indirect contribution leading to the successful completion of this research.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
ML	Machine Learning
RF	Random Forest
XGBoost	Extreme Gradient Boosting
ETU	Emergency Treatment Unit
API	Application Programming Interface
UI	User Interface
DB	Database
MAE	Mean Absolute Error
RMSE	Root Mean Square Error
WAPE	Weighted Absolute Percentage Error
NLP	Natural Language Processing
GDP	Gross Domestic Product
NoSQL	Not Only Structured Query Language
JS	JavaScript
JSON	JavaScript Object Notation
CRUD	Create, Read, Update, Delete
HTTP	HyperText Transfer Protocol
CPU	Central Processing Unit
RAM	Random Access Memory

1. INTRODUCTION

1.1. General Introduction

The healthcare domain is instrumental in ensuring the well-being and sustainability of societies. On the other hand, in the modern world with a demand for medical needs, limited resources availability and complexity in management of dynamic and non-determined patients' inflow, health care systems become more demanding. A healthcare system, more so a hospital, especially if they are in a developing country consistently fails to maintain an optimum approach between resources available and patient need s this has led to operational wastages at the other end and reduced quality of care.

The successful use of pharmaceutical resources is one of the difficult challenges in hospital management. Medicines are vital components of health care delivery system; in ETUs, immediate accessibility of medicines may often influence the survival and recovery of patients. But what drives the complexity of having a good medicine inventory is dynamic demand patterns, seasonal aspects and sudden emergency events including road traffic accidents.

Though, the hospitals have to rely on manual estimation methods and historical averages for determining medicines needs. These approaches might even help give a rough idea of inventory management but are more or less limited in adjusting to real-time fluctuations and unexpected spikes. Therefore, hospitals often deal with problems like medicine shortages during peak demand and overstock during low-demand cycles. While all of this entails extra operating costs, the unrealistic expenditures also create wastage through medical resources paying for high end equipment and also create marginality in the quality of patient care.

By exposing the power of data analytics and machine learning techniques in recent years, we believe that these new tools are a step forward to overcome these challenges. Machine learning models can sift through huge amounts of historical data, finding untapped patterns, and make accurate forecasts about future demand. These competencies allow healthcare systems to shift from reactive decision-making approaches towards proactive and predictive management strategies.

External factors such as road traffic accidents are especially important for healthcare demand. Accident-related injuries are a major cause of emergency hospital admissions in some countries, such as Sri Lanka. Such occurrences often happen randomly, with a sudden influx of patients and hospitals are pressed for medicine. Not being able to foresee such spikes adds up to the challenges faced in inventory management as well as emergency preparedness.

Ensuring availability of medicines receives greater attention due to this research on its part addressing the above challenges by developing a machine learning–based system for demand forecasting and accident surge detection. The proposed solution incorporates prediction analytics methods to estimate the consumption of regular medicines and simultaneously detect future rises in demand for accident-related emergencies. With all these two ingredients, this system approaches a more holistic and intelligent way.

The study uses data from historical medicine usage records and accident admission recordings from Yakgahapitiya Divisional Hospital. With these datasets, machine learning models like Random Forest and XGBoost are used to capture complex patterns and relationships in the data. The system automatically and dynamically adjusts medicine demand predictions on identification of the surge condition which enhances the accuracy and reliability.

Additionally, the designed system is realized as a web-based application enabling healthcare administrators to interpret predictions and make effective decisions of inventory management. By combining predictive modeling with actual implementation of the systems, this research guarantees that its implications span beyond the theoretical realm and also practice.

Therefore, providing medical demand forecasting system, with a data-driven and intelligent approach can improve the patient flow management which enhances the usability of hospital resources. The system, which uses machine learning techniques along with accident surge detection in hospitals to close long-standing gaps in traditional hospital management practices, will also help improve efficiencies, minimize wastage and make essential medicines available at the right time.

1.2. Background literature

With the continuous increase in demand for medical services and the increasing complexity of hospital logistics operations, deciding on efficient healthcare resources management especially about pharmaceutical supplies is a major research field. The combination of data analytics and machine learning on healthcare systems offer novel ways to achieve better forecasting, resource utilization, and decisions. The remainder of the paper provides a review of the relevant literature in medicine demand forecasting, healthcare supply chain management, accident analysis and machine learning in healthcare.

- Pharmaceutical Supply Chain and Availability of Medicines

Access to the essential medicines is a key component required for efficient healthcare systems. The pharmaceutical supply chain based in developing countries experience lack of infrastructure, limited forecasting methods and coordination

among supply chain stakeholders could lead to not maintaining a steady supply chain.

Ranasinghe et al. highlighted that accurate demand forecasting and effective inventory management can help improve hospital supply chain performance. According to the study conducted by them, estimation of need for medicine in case of poor results will lead to excess supply or a shortage. These inefficiencies lead to poor patient care and high operational costs.

Similarly, Buddhika et al. analysis of availability of medicine in Sri Lanka and reported that the availability essential medicines were >80% beyond WHO benchmark in many instances but some drugs still faced stock-out period. Misestimating demand and lack of coordination between SC entities were found to be major causes of these shortages as per the study. This suggests that hitting general availability targets does not equal assured access to all the medicines you need. The findings in this paper indicate that while some progress has been made to the distribution systems of pharmaceutical, improved forecasting methodologies are still very much needed for ensuring continuous and reliable availability of medicine.

- Demand and Supply Factors in Healthcare: Technological Advances in Healthcare Delivery Systems

In an attempt to overcome the inefficiencies of conventional supply chain management, various technology applications have been introduced in healthcare systems. SWASTHA system developed by the Ministry of Health is one such example in Sri Lanka.

SWASTHA was introduced by Attanayake and Rambukwelle as a web-based platform to automate the major functions such as demand estimation, procurement, distribution and inventory tracking. This system is designed to reduce manual interventions and to increase coordination among different stakeholders in order to bring more transparency and efficiency into pharmaceutical supply chain activities.

Though SWASTHA has come a long way from traditional digital health systems, it is primarily an operational management tool — not one with predictive intelligence. Advanced machine learning methods are not used to predict future demand or identify unusual spikes in the use of medications. Thus, it does not have the ability to respond reactively whenever there is a change in health care demand.

This criticism fails to recognize the importance of embedding predictive analytics alongside current systems within our healthcare design, such that it can enable the systems and adapt.

- Effects of accident claims admission on health service demand

Road safety continues to be an issue, particularly in developing countries where accident-related incidents are the biggest cause of emergency hospital admissions. These events frequently cause surges in patient volume, creating immense stress on hospital capacity.

Kodithuwakku and Peiris are studied about road traffic accidents in Sri Lanka and mentioned key factors which report accidents. This includes environmental conditions (e.g., weather, road quality), temporal factors (e.g., weekends, nighttime) and human factors (e.g. driver behavior). It emphasized the fact that accident patterns are not random; rather they are influenced by identifiable trends and conditions.
Results indicate that historical data and pattern recognition techniques could help a predict accident-driven admissions to a certain extent. Nevertheless, current studies use either existing information to examine causes of accidents rather than integrating this knowledge into healthcare planning.

This gap can be characterized by limited integration of accident analysis, and the management of hospital resources for emergency preparedness. Accident surges are something hospitals respond to reactively instead of proactively -- resulting in supply chain and resource allocation inefficiencies.

- Machine Learning Applications in Forecasting in Healthcare

Among many other tasks, machine learning has shined in healthcare because it can analyze large datasets and provide accurate predictions. Machine learning techniques have been used with various degrees of success for tasks ranging from disease prediction, patient monitoring to resource forecasting.

Random Forest is a supervised, ensemble learning method for classification and regression tasks. It is an ensemble learning method which builds multiple decision trees and merges them together to get a more accurate prediction. It works particularly well on non-linear relationships and complex datasets, which makes it a good fit to forecast a medicine demand based on usage history.

XGBoost (Extreme Gradient Boosting) Machine learning algorithms help machine capture knowledge and represent it in the context standard style. They perform a boosting mechanism to improve the model predictions by reducing mistakes sequentially. XGBoost is efficient and scalable, making it ideal for cases such as ADMS (accident detection management system) where patterns can work out to be sudden or non-persistent leading to high output.

These models have shown high effectiveness in several healthcare applications or studies. Systems that predict demand using machine learning have been more accurate than traditional statistical forecasting approaches, allowing for better planning and resource allocation. Most of these studies targeted single domain applications, such as the ability to predict patient admissions or disease trends [13], but unified system modelling is usually not adopted.

- Limitations of Existing Approaches

However, existing management practices related to hospital resources still have many barriers despite the rise of healthcare technology and machine learning causal applications. In the first instance, most systems utilize simplistic forecasting approaches based on previous averages, which do not take into account live variations in demand. These approaches cannot detect rapid changes brought by emergency events, thus resulting in imprecise predictions.

Moreover, the challenges of forecasting medicine demand and the analysis of accidents are generally studied independently (8). Studies of pharmaceuticals generally follow the building blocks of inventory management, whereas accident studies tend to identify causes and patterns. The separation between these domains makes decision making less effective in health care systems.

Third, manual processes are still predominant for resource planning among healthcare institutions particularly in developing countries. They tend to be slow, error-prone, and are not ideal for processing large volumes of data. Third, established digital systems like SWASTHA do not include sophisticated predictive models and therefore are unable to offer actionable decision support.

- Contribution of the Proposed Research

The identified limitations with the work done previously are addressed in the proposed research by modelling an integrated framework using machine learning for medicine demand forecasting and detection of accident surges. This research links two important aspects of healthcare demand that previously have been modelled separately: the normal routine consumption of medicine, and conditions leading to emergency surges in the provisioning of services, which current systems do not account for. A Random Forest Model enables to predict normal medicine demand accurately, while XGBoost model is used for detection of any potential high demand due to accident triggered incidents.

Using the combination of these models, the system adapts its predictions in real-time based on current conditions, making it a more resilient and adaptive way to manage resources. It allows hospitals to predict demand better, ensure fewer medicine shortages and combination of medicines for emergencies. In addition, using the system as a web-based application makes it usable in practice and ensures that

healthcare administrators can get predictions quickly from the system to help HIM take data-driven decisions.

Overall, this study advances the management of healthcare resources by demonstrating a holistic approach to predictive analytics, machine learning and system implementation. This identifies significant gaps in the literature and also presents a scalable framework for enhancing hospital operations and patient experience.

1.3. Research gap

No related review has been identified thus far that consolidates the literature on pharmaceutical supply chain management, accident analysis and machine learning applications in the healthcare domain leading us to believe that considerable developments have taken place in each of these domains independently. For applied integrated hospital resource management systems, however, some crucial gaps nevertheless exist.

At first, most of the studies addressing pharmaceutical supply chain management have focused their attention towards improving access to medicines through appropriate inventory levels and distribution mechanisms. For instance, systems like SWASTHA have truly automated many procurement and inventory processes, leading to better transparency and operational efficiency. But, demand planning for these systems is based on historical data and manual calculations. They do not have high-level prediction ability to predict accurately health futures and accept that most things, especially in complex dynamic rapidly changing conditions such as public medicine will be wrong. This limitation undermines their competence in managing hospital requirements with real-time fluctuations.

Second, studies from the perspective of accident analysis mainly deal with the causes, patterns and risk factors of road traffic accidents. Although these studies offer important insights regarding trends in accidents, they do not take their application to the level of healthcare resource planning. The lack of linkage between accident prediction and forecasting demand for hospitals would lead healthcare systems to prepare for emergencies on a reactive basis [17]. Consequently, hospitals are usually driven to reactively respond to sudden spikes inpatient admissions.

Thrid, the machine learning techniques such as Random Forest and XGBoost have been largely done for predictive analysis in healthcare [8], but the several applications are generally confined to a single domain problem. Previous studies have either predicted the admission of patients (clinical and practical use) or prediction of type of diseases but using single factors only instead of multiple interdependent influential components which lead to demand for hospital resources.

It is this absence of a multi-dimensional approach that limits the capabilities of machine learning, especially so far as complete decision support goes.

The other major gap is fewer decision-making systems driven by real-time data in healthcare environments, especially the setting of low-income countries. Most hospitals continue to rely on manual processes and static models for resource planning that are not able to withstand sudden changes, like an emergency incident or seasonal trends. This leads to unnecessary resource allocation, increased operational costs and reduced quality of patient care.

In addition, current systems cannot well accommodate the association between usual consumption of medicine and demand shocks due to emergencies. In practice, the usage of medicine is determined by both normal and emergency patient treatment for example due to accidents. Predicting these two factors combined in a single forecasting model is tough, which results in inaccurate forecasting and inefficient inventory management.

In summary, the major research gaps highlighted are as follows:

- Absence of predictive knowledge in the present-day pharmaceutical supply chain systems
- Lack of integration of accident analysis with health care resource allocation and planning
- Lack of usage of multi-factor machine learning models in hospital demand forecasting
- Reliance on historical and manual forecasting
- Dynamic demand of medicine cannot be altered under emergency situations

This indicates a clear necessity for a single integrated machine learning based system possibility considering both the forecasting of demand for medicines as well as the accident surge detection. With this kind of system, hospitals could advance from a reactive resource management practice to a proactive one, allowing for better forecasting, emergency preparedness and overall healthcare operations.

Table 1 : Comparison of Existing Systems vs Proposed System

Study /System	Focus Area	Use of Machine Learning	Demand Forecasting Capability	Accident/ External Factor Consideration	Real-Time Adaptability	Integration of Multiple Factors
Ranasinghe et al. [11]	Drug supply chain management	No	Limited (manual estimation)	No	No	No
Attanayake & Rambukwelle (SWASTHA) [12]	Medical supply automation system	No	Basic estimation (not predictive)	No	Limited	No
Buddhika et al. [13]	Medicine availability & stock-out analysis	No	No	No	No	No
Kodithuwakku & Peiris [14]	Road traffic accident analysis	Partial (statistical models)	No	Yes	No	No
Proposed System	Medicine demand forecasting + accident surge detection	Yes (Random Forest, XGBoost)	Yes (high accuracy prediction)	Yes (accident surge detection)	Yes	Yes

As displayed in Table 1.3.1, the comparisons show that the literature-oriented studies and systems have focused on a single area of healthcare management a supply chain operations or accident analysis. From the systematic analysis of existing methodologies none of reviewed approaches integrate predictive medicine demand forecasting with accident surge detection. It overcomes these constraints through the use of machine learning components to mine past medicine usage and accident records. It is unlike the basic systems it integrates with and provides real-time adaptability while enabling decision making via multi-factor integration in the healthcare establishment.

1.4. Research Problem

Medical resources management is a central problem in health institutions, especially when the environment has limited medical resources and high demand from patients with fluctuating risks. Access to essential medicines is one of the many resources necessary for running a hospital and therefore a composite, vital piece in delivery speedy and effective management of patient care. But it is not easy for hospitals to keep the medicine supply and demand ideally balanced because healthcare situations are so unpredictable.

Traditional forecasting methods are a key reason behind this problem area. The average number of patients lying down is all the hospital has seen so far for years or possibly decades, as well as man-made estimates about which medicines will be needed. Although these methods are basic approaches toward planning, they fail to adapt in real-time variations and spikes of demand. Forecasting errors consequently result in overstock or stockouts. Having too much medicine means high storage costs and wasted drugs, particularly short expiry medications, while shortages push back therapy initiation in patients and risk their safety.

In emergency units you can easily see that if the medicine requirement can fluctuate suddenly, for example due to an unexpected event such as a road traffic accident. Emergency cases due to accidents amount major share of the in admissions like in Sri Lanka. Such incidents often happen unexpectedly and can lead to cases rushing into the hospital which in turn puts a lot of pressure on the resources of the hospital. For such events, demand for specific emergency medicines surges over a short time period in a manner that hospitals simply cannot respond to effectively using standard planning tools.

Furthermore, existing hospital management systems are not designed to integrate multiple data sources for decision-making. Medicine usage data and accident-related admission data are typically analyzed independently, without considering the relationship between routine consumption and emergency-driven demand. This lack of integration prevents healthcare administrators from obtaining a comprehensive understanding of demand patterns and limits their ability to make proactive decisions.

Third critical issue is that many health care organizations do not yet have functional real-time decision support systems. And manual processes are not only time-consuming but very human error-prone, lowering the reliability of planning and resource allocation. Furthermore, the limitation of the methods ability to process a large amount of historical data highlights a further limitation.
Such challenges stress the need for a more advanced an inspired hospital resource management. A system that is capable of forecasting medicine demands efficiently whilst monitoring spontaneous surges due to accidents requiring emergency

medicines. This type of system would allow hospitals to better predict demand variation, decrease levels of stock needed on hand, and increase preparedness in emergencies.

As a result, the research problem tackled in this study can be described as follows:

What features can an integrated machine learning–based system be developed to correctly estimate medicine demand and identify accident-related spikes, thereby improving hospital resource management and making sure essential medicines are available in time?

1.5. Research Objectives

The primary aim of this research is to address the challenges associated with inefficient medicine inventory management and the lack of predictive mechanisms in hospital systems. By leveraging machine learning techniques and data-driven approaches, this study seeks to develop an intelligent solution that enhances the accuracy of demand forecasting and improves emergency preparedness in healthcare environments.

The objectives of this research are formulated to ensure a systematic approach to analyzing the problem, designing predictive models, and implementing a practical system that can support decision-making in hospital operations.

1.5.1. Main Objectives

This research primarily aims to design an integrated ML–based demand forecasting and accident surge detection system for medicines, in order to optimize hospital resources management through improved emergency preparedness and proactive decision-making in healthcare.

1.5.2. Specific Objective

The research objectives, offered in a systematic manner, are intended to tackle the challenges around resources management in hospitals and to facilitate successful development of an integrated predictive system. Objectives: First, to evaluate historical medicine use data and identify effects on an underlying trend of pharmaceutical consumption in a hospital environment. Analyzing this is the basis to build reliable forecasting models.

Second, the study presents the design of a routine medicine demand prediction machine learning model based on Random Forest during normal operation period. It is assumed that this model will capture complex, non-linear relationships in the data

and predict future medicines and foresee reliability on a self-sufficient period-of-time basis.

Furthermore, the study will develop an XGBoost based model to identify excess hospital visits attributed to accidents. This element is central to overriding short spikes in emergency cases that more than trend the requirement of medicine. The model aims to identify accident-related patterns during high-risk periods and generate alerts in advance by analyzing related data.

A second primary goal is to link both forecasting and surge detection methods under a single system. This integration allows predicted medicine demand to be rapidly updated according to current conditions, ensuring that routine and emergency-induced fluctuations in demand are captured within the system.

Additionally, the study aims to assess the performance of created models based on common quality metrics (MAE, RMSE, WAPE) and classification-oriented metrics (accuracy, precision, recall & F1). This is to evaluate the adequacy of the proposed system.

Finally, this project involves the deployment of the developed models on a web application using Flask where an intuitive interface and user experience provide visualizations of predictions to aid healthcare administrators in decision-making with respect to medicine inventory management. This pragmatic approach ensures that the research can be used not only in theory, but also for improving healthcare.

2. METHODOLOGY

2.1. Introduction

This chapter describes the plan adopted to design, implement and assess the proposed ML-based medicine demand forecasting and accident surge detection system. The methodology of the study was planned such that it guarantees achievement of research objectives in a systematic way and also enables practical applicability in healthcare settings.

This research mainly focuses on the bottlenecks in medicine inventory management and prediction problems in hospital systems. This employed a combination of data analysis and machine learning model development approach, with system implementation techniques to achieve this. The methodology combines both analytical and solution-oriented aspects to ensure the provided solution is both theoretically valid and practically applicable.

The primary research method adopted was a quantitative approach. Utilizing this approach is particularly well-suited due to the structured nature of the underlying datasets including historical records for medicine use and admission data related to accidents. From these dataset's patterns, trends and relationships affecting medicine demand within the hospital environments were analyzed. Various machine learning techniques and statistical analysis were employed to convert raw data into results and predictive outputs.

The developed system is made up of two main functional parts. First, the medicine demand forecasting, which is a Random Forest regression model predicting average consumption of routine medicines based on non-pandemic scenarios. Second, the accident surge detection module uses an XGBoost model to determine underlying surges of accident-related admissions that could lead to sharp increases in demand for medicines. These two parts are combined to form a complete and adaptive forecast mechanism.

Apart from development of models, methodology also covers design and implementation of a web-based application involving the implementation of predictive models in a user-friendly interface. The system allows the healthcare administrator to enter data, view predictions and make data-driven decisions when it comes to either managing inventory or distributing resources. This way, by combining machine learning models with an application in the real world, work can meaningfully add to academic knowledge with an emphasis on solving problems.
In addition, this methodology includes model validation methods which evaluate the performance and reliability of the developed system. To measure the forecasting model, we used standard evaluation metrics like Mean Absolute error (MAE), Root Mean Square Error (RMSE) and Weighted Absolute Percentage Error (WAPE). For

the accident surge detection model classification metrics such as accuracy, precision, recall and f1-score were used.

Detailed descriptions of the research design, data collection, preprocessing methods, model development processes etc. system architecture and testing procedures are presented in subsequent sections of this chapter. It is through this structured approach, that the paper hopes to contribute to an effective, scalable and intelligent system for hospital resources management and emergency preparedness.

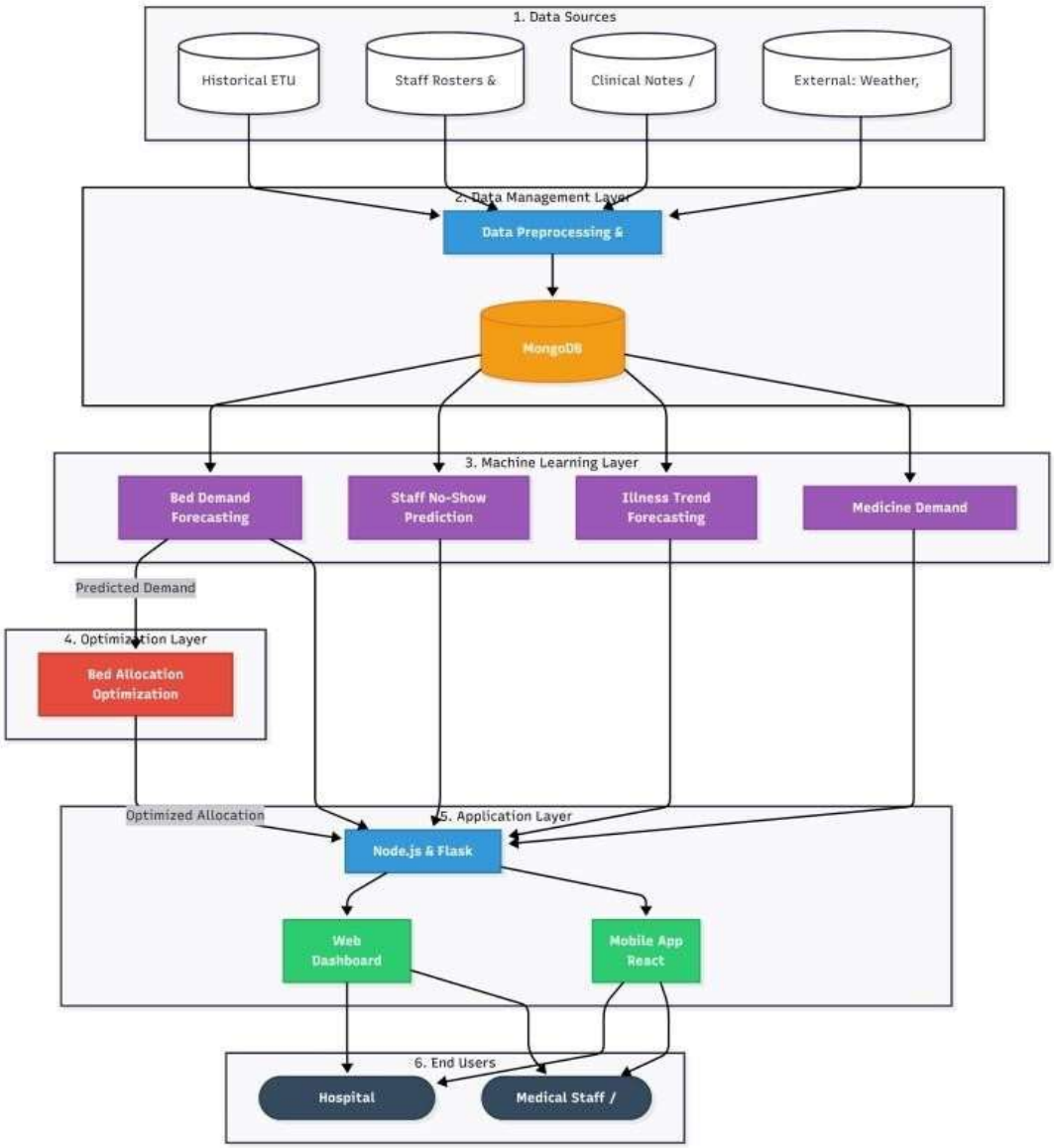

Figure 1: System Architecture Diagram

2.2. Research Design

This study uses a data-driven and experimental research design in order to develop and validate a machine learning–based system capable of medicine demand forecasting and accident surge detection. The design is laid out in a manner that shows a clear thought process — starting from selection of the problem, to solutions and their verification which correspond to the overall aim of betterment of hospital resources.

We will also effort to incorporate into the study factors such; gross domestic product, national drug control policy, strengthen and improve the health delivery system (more especially pharmacy laboratory) to assess any complementarity in demand for the hands-on nature of medicine based on associated services. The study is mainly targeted towards developing countries where medicine is virtually provided based on affordability through out-of-pocket payments subjected within individual scopes with historical healthcare data pattern those define relationship between personal and interpersonal influence for medicine-derived demand. We therefore adopt a quantitative research methodology because such an approach allows investigation with structured datasets, and the use of statistical and machine learning techniques. It guarantees that the developed models are data-driven and generated measurable and reproducible results.

The research design contains some significant phases relate to the data collection, processing of data, extraction of features, development of model, predictive components integrations and evaluation of performance. All phases are carefully crafted to serve the end effectiveness of the suggested system.

First, relevant datasets are collected from Yakgahapitiya Divisional Hospital such as the records of using the medicine and admissions due to accidents. These datasets serve as the basis for demand insights and drivers of medicine use in the population.

Data preprocessing techniques were applied to clean and prepare the data for analysis after data collection. This included dealing with missing values, eliminating inconsistencies in the data and converting categorical variables to a numerical format that can be processed by machine learning models. Feature engineering was conducted to derive useful features like time-related features and usage trends, that may be informative with respect to the predictions of the models.

At the center of the overall research design is creating two machine learning models. The first model, a Random Forest based one, is used for predicting demand of routine medicine under normal conditions. For such future increases in accident-related admissions, a second model based on XGBoost is developed specifically for identifying potential cases. These models are built using past data, and validated against common performance measures to ensure their accuracy.

An important element of the research design is a merger of these two models into one system. The system intends to change automatically statistics of medicine demand forecasts as surge conditions are detected, making it more realistic forecasting. By overcoming traditional single-model limitations, it contributes to an efficient solution package. The research design consists of both model development and implementation of a web-based system for users to interact with the predictive models. This system can be used effectively by health care administrators for inputting data, visualization of predictions and demand trend monitoring.

Finally, the performance of the system is studied using regression and classification metrics to evaluate how well it performs in real life. This approach supports evaluation of the proposed solution to determine whether it is accurate and reliable enough for deployment in healthcare settings.

In fact, all aspects of the research design form a complete framework for designing an intelligent and integrated hospital resource optimization system. Integrating data analysis, machine learning and system implementation as well as providing a solution that would be robust and scalable enough to tackle the issues identified in previous sections, are all dimensions addressed by the study.

2.2.1. Requirement Analysis

The requirement analysis phase is an important heading of the research design that which specifies and draws out the functional as well as non-functional needs of the intended system. This step of understanding lived outcomes, stakeholder needs (similar to the stages above) and mapping out high-level specs of what a system would need to do in order to fix these issues. The goal is to ensure that the implemented solution can solve real-life problems associated with medicine demand forecasting and emergency preparedness.

Structured approach to collect and analyze requirements both tech and domain perspective. These requirements were derived from the availability of data, current healthcare workflows for prevention protocols that could be enhanced through a predictive system.

2.2.1.1. Stakeholder Identification

Realization of the proposed system is contingent on gaining insights into stakeholder expectations and needs relevant to hospital operations. In this research, the main stakeholders are identified as follows:

- Hospital Administrators

Hospital administrators must manage resources: what type of medicine to keep, inventory and buy processes. They need demand forecasting data, and timely insights into stock management, budgeting, and resource allocation in order to make strategic decisions.

- Medical Officers ETU Staff

The availability of essential medicines is between their timely treatment by doctors, ETU (emergencytreatment unit) staff. Their key responsibility is to keep the medication inventory at an all-time high during normal functioning and unprecedented situation needs (like accident surges).

- Pharmacy Staff

Also, Pharmacy personnel does all the work related to handling of medicine including inventory control, usage tracking and maintaining stock levels. They need a system which offers predictive demand and timely notice to prevent from running out of stock and make it efficient in eliminating wastage.

- Data Providers (Hospital Records)

The system relies on historical data collected from hospital records, including medicine usage logs and accident admission data. The quality and accuracy of this data are crucial for developing effective machine learning models.

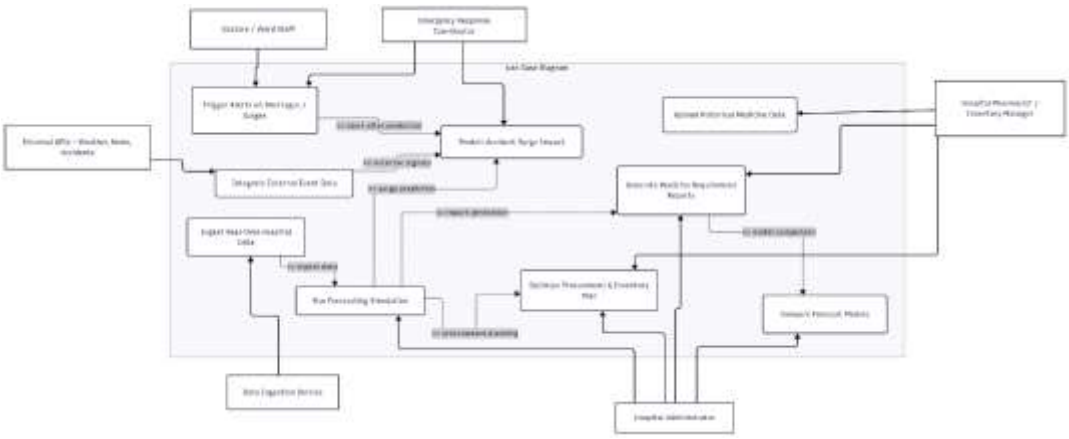

Figure 2 : Use Case Diagram

2.2.1.2. Data Requirements

The success of the machine learning system that is suggested depends of heavily on the input datasets respectively quality, structure and completeness level. Two main structured datasets were used in this work which were both gathered from the Yakgahapitiya Divisional Hospital in collaboration with medical personnel. These datasets are chosen to model average hospital operation conditions and emergency situations so that an integrated forecasting system can be developed.

Figure 3 : Yakgahapitiya Divisional Hospital

- Medicine Usage Data

The first dataset is daily medicine information obtained from hospital pharmacy systems and Emergency Treatment Unit (ETU) logs. This dataset has around 16500 records over a period of time from 2022 to 2026, making it relatively complete in terms of the historical medicine consumption profiles. The dataset consists of the following features for each row:

- Date - the day on which the medicine was issued or played out
- Medicine_Name - The Name of Medicine administered.
- Qty_Used – Quantity of the medicine for that day

It started with 11 essential medicines most often used in a hospital environment. A set of four most used ETU medicines were chosen for this study to train the forecasting model. The above medications were selected for readiness based on high frequency and clinical relevance to the emergency management setting.

The dataset is fundamental in effort of seeking daily change, weekly trend, and seasonal variation in scope of medicine consumption. From these patterns the system can accurately predict future medicine requirements under normal operating conditions. The mentions in the dataset enriches and lengthens the amount of time we can rely on forecasting model outputs.

- Accident Admission Data

Hospital admission records related to accident, which is needed to accommodate all emergency-driven changes in medicine demand. It is composed of roughly 10,000 records on emergency admissions in the corresponding timeframe. This dataset consists of records with the following attributes:

  - Date — They date when admitted to the hospital after an accident.
  - Accident_Type – Type of accident (road traffic accidents, falls etc.)
  - No_of_Cases – A count of patients admitted from the particular accident type
  - Injured – The kind of any injury incurred
  - Medicines_Used – The medications provided for treatment

This dataset has four types of accident categories, and we trained the accident surge detection model on all. This data set allows access to patterns and trends linked to the occurrence of accidents such as time periods like a special season or weekends jut cylinders where there are more incidents.

With the help of this data, the system can identify abnormal rises in accident-related admissions--known as howling alerts of impending tsunami conditions. Fully aware of these surge conditions which play a crucial role in determining demand for medicine, especially at emergency units where every second is precious.

- Data Quality and Importance

It is crucial that both datasets (test and train) are consistent, comprehensive and are in a properly formatted way for Machine learning models to succeed. Data Cleaning To guarantee data quality, we used several data cleaning steps like missing management, managing inconsistencies and the formatting of standardizations.

Between these two datasets, we can obtain a more complete picture of hospital operations by picking up changes based on both demands stemmed from routine medicine consumption and demand fluctuations driven by emergency. This two-input strategy allows for creating a comprehensive system that can generate more precise and flexible awareness in predictions compared to conventional approaches using one data entry only. As a whole, the availability of a discrete and high-quality dataset improves predictive models which guarantees the reliability of proposed system in actual healthcare scenarios.

2.2.1.3. Functional Requirements

Functional requirements describe what the system must doing in order to fulfill objectives it is meant to achieve. These specifications can precisely depict how a system works, how it interacts with data and users. This research presents a system that aims to facilitate data-driven decision-making in the hospital, producing accurate forecasts for medicine demands and detecting conditions of emergency surge.

The following functional requirements were identified to achieve this:

- The system shall allow the input, storage, and retrieval of historical medicine usage data from hospital records.
- The system shall allow the input and processing of accident-related admission data to analyze emergency trends.
- The system shall preprocess and clean raw datasets to ensure data quality before model training.
- The system shall generate predictions of medicine demand using a Random Forest machine learning model.
- The system shall analyze accident data using an XGBoost model to detect potential surge conditions.
- The system shall dynamically adjust predicted medicine demand based on detected accident surges.
- The system shall provide real-time or near real-time prediction results to users.
- The system shall display prediction outputs in both numerical and graphical formats for easier interpretation.
- The system shall provide alerts or notifications when abnormal demand patterns or surge conditions are detected.
- The system shall allow users to compare historical data with predicted values for analysis and validation.
- The system shall support updating datasets periodically to improve model accuracy over time.
- The system shall maintain a history of predictions and results for future reference and analysis.

These functional requirements ensures that the system is capable of performing both routine forecasting and emergency detection thus offering a holistic solution for hospital resource management.

2.2.1.4. Non-Functional Requirements

Quality attribute, performance criteria and operational constraints for the system that are defined as non-functional requirements. This ensures that the system is robust, efficient, and capable of being deployed in real-life situations from a clinical perspective.

For the proposed system, the following non-functional requirements were identified:

- Accuracy - Ensure that the system makes highly accurate predictions to allow for beneficial decision-making. Cause to minimize the prediction errors optimized machine learning models and evaluation techniques.
- Performance - The system must efficiently handle large datasets and produce predictions within reasonable periods. User requests should have a very short response time in order to operate seamlessly.
- Scalability - The system should be able to deal with growing volumes of data as the records at hospitals grow over time. Be expandable into the future (i.e., more data and/or hospital systems).
- Usability - The user interface must be clean and straightforward for even a novice able to navigate the system. User experience should be enhanced with clear visualizations and straightforward interactions.
- Proficient - The system should work reliably without breakdowns or lapses. Should respond to errors in a controlled manner and make sure services are available all the time.
- Security - Healthcare data is sensitive in nature and needs to be protected against unauthorized access. Proper safeguards must be in place to control access (e.g., use authentication) and ensure only the authorized data is properly handled.
- Maintainability - A more modular system is good so that any update and debugging or improvements can be made without affecting any other part of the systems functionality.
- Flexibility - The system needs to be adaptable in different hospital settings and able to integrate additional data sources, or models as needed.
- Data Integrity - The system shall hold and process the data correctly without losing or corrupting any such data.

These functional requirements allow the system to not only maintain its intended functions but more importantly, preserve a high standard of quality and reliability in a health care setting by ensuring certain non-functional requirements.

2.2.1.5. System Requirements

System requirements specify the technical infrastructure required to develop, deploy and run the proposed solution. These consist of both hardware and software components required for the operation of systems.

Hardware Requirements

- Standard for dev/test — standard compute (desktop or laptop)
- Minimum 8 GB RAM to work efficiently with data and to train your model
- At least enough storage space to be able to store the datasets and system files
- Server / Cloud based infrastructure deployment (optional for scalability)

Software Requirements

- OS - Windows, Linux or macOS
- Programming Languages - JavaScript, Python
- Frontend Framework - React. So just using this api with redux and react js for frontend.
- Backend Framework - Node. Server-side logic with js and Express.
- Database Data - Mongo DB for structured and semi-structured data
- Machine Learning Libraries
    - Scikit-learn library for Random Forests
    - Using the XGBoost Algorithm for detecting accident surge
    - Pandas and NumPy for processing data
- Visualization Tools - The tools for visualizing the data, Matplotlib or a similar library that generates graphs and charts.

Integration Requirements

- PyNode - Communication between the Python-based machine learning models and the Node.js backend
- Data exchange from front-end to back-end via API
- Connection to a so-called Secure database or ability to store and extract data

There are the system requirements which make sure that the application is developed on a strong and scalable technological basis.

2.2.1.6. Requirement Analysis Results

We used the requirement analysis phase to better understand about needs and problems regarding hospital resource management. Several key findings resulting from an analysis of stakeholder needs, data availability and system limitations. The number one is the need for moving from manual forecasting to predictive but data-

drive system. Stakeholders stressed the value of forecasting for better management of inventories and reduced medicine shortages.
Secondly, the need to integrate various data sources proved to be an important functional need. Having both information on how much medicine is being used, and also the admission-related data for accidents makes it possible for the system to understand changes in demand that are truly pushing demand versus adopting normal patterns of activities, which results in higher fidelity predictions and better forecasting.

Third, the need for real time or near real-time decision support was emphasized. Healthcare administrators need timely insights to be able to respond appropriately for a sudden change of demand, especially in a crisis. A mean entry way was also found in usability and simplicity as facilitators of adopting the system. Outputs and visualizations must be unique to the non-technical user so that they can digest key findings without needing technical assistance.

Lastly, requirement analysis established the need for methods to process complex patterns in data along with accurate forecasts by applying machine learning techniques. Applying models like Random Forest and XGBoost caters to these needs, at the same time be sure it performs further up in performances.

In summary, the results from phase 1 establish a basis for design and development of the system. They ensure that the solution you propose is relevant to what health care actually needs and no other approaches work better.

2.3. Application Development Process

To develop the proposed system, a structured and modular application development process was adopted to ensure accuracy, scalability, and effective integration of machine learning models within a real-world healthcare environment. The primary objective of this process was to transform the conceptual framework of medicine demand forecasting and accident surge detection into a fully functional and operational system capable of supporting decision-making in hospital resource management.

The development lifecycle consists of several well-defined stages, including the selection of appropriate tools and technologies, system design, model development, model integration, and system implementation. Each stage was carefully aligned with the functional and non-functional requirements identified during the requirement analysis phase. This structured approach ensures that the system is both technically robust and practically applicable in real healthcare scenarios.

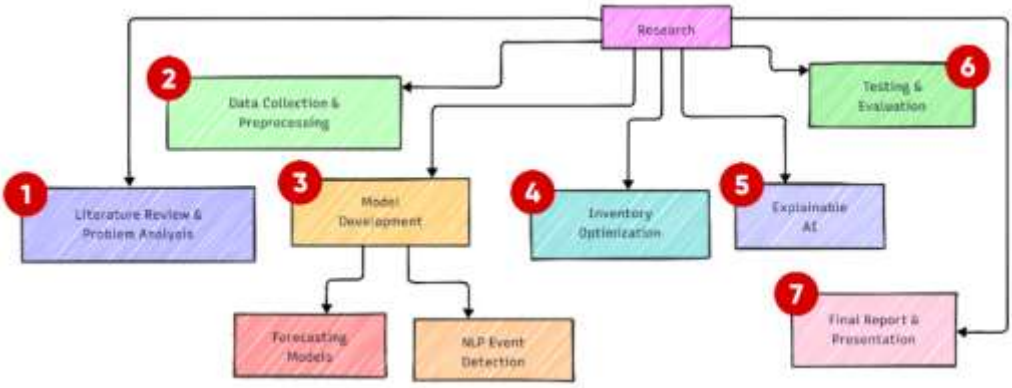

Figure 4 : Work Breakdown Structure

2.3.1. Tools And Technologies

- Frontend Development – React.js

The frontend of the application was developed using React.js, a declarative and component-based JavaScript library widely used for building interactive user interfaces. React enables efficient rendering through its Virtual DOM mechanism, which improves performance when handling dynamic data updates.

React was selected due to its ability to support real-time updates and reusable component structures. This allows the system to efficiently display changing prediction values and visualization outputs. The frontend is responsible for:

- Plot predicted medicine demand values
- Using charts and graphs to visualize trends
- Alerting for mutual conditions
- Allowing users to interact with the system

Built using React for a fast and interactive user experience, making it extremely easy for healthcare administrators.

- Backend Development – Node. js with Express

We built the backend of the system using Node. js and the Express framework, which is a minimal web application framework for node. Among other things, backend performs several critical functions:

- Front Ent API Requests Handling
- Processing input data
- Communicating with the database
- Integrating machine learning predictions
- Control of system logic and workflows

The asynchronous nature of Node. In this case, the js will enable to handle many requests so it is able for real time apps.

- Database Management – MongoDB

The system utilizes a NoSQL database (mongoDB) for storage and management purposes. They went with MongoDB due to the flexibility it needed for handling both structured and semi-structured data along with being optioned scalable for large datasets. The database stores:

  - Historical medicine usage data
  - Accident admission records
  - Outputs of model & prediction result

MongoDB provides efficient data retrieval and supports scalability, allowing the system to handle increasing volumes of hospital data over time. Its document-based structure is particularly useful for storing complex datasets without requiring rigid schemas.

- Machine Learning Development — Python Environment

We had a machine learning component in the system which we implemented using Python, as Python has a mature ecosystem for data analysis and predictive modeling. The following libraries were used:

  - SciKit Learn for No of Random Forest members!
  - XGBoost to detect surges in accidents
  - Pandas and NumPy for data manipulation and preprocessing
  - Matplotlib and Seaborn Visualization tools and data analysis

Python has been chosen because it is simple, well-known libraries and can-do wonders with large dataset.

2.3.2. Design

To achieve separation of concerns and ease of maintenance, a modular architecture was used for the design of the system. This emphasizes the integration of several components into one framework that allows for data processing, prediction and visualization. Our system architecture is comprised of four layers:

- Data layer – The heavy lifters that store and manage the datasets
- Processing Layer – Meant to perform data cleaning and preprocessing
- Churn Detection and Machine learning layer: Prediction and surge detection
- Application Layer — Interaction between frontend and backend for the user.

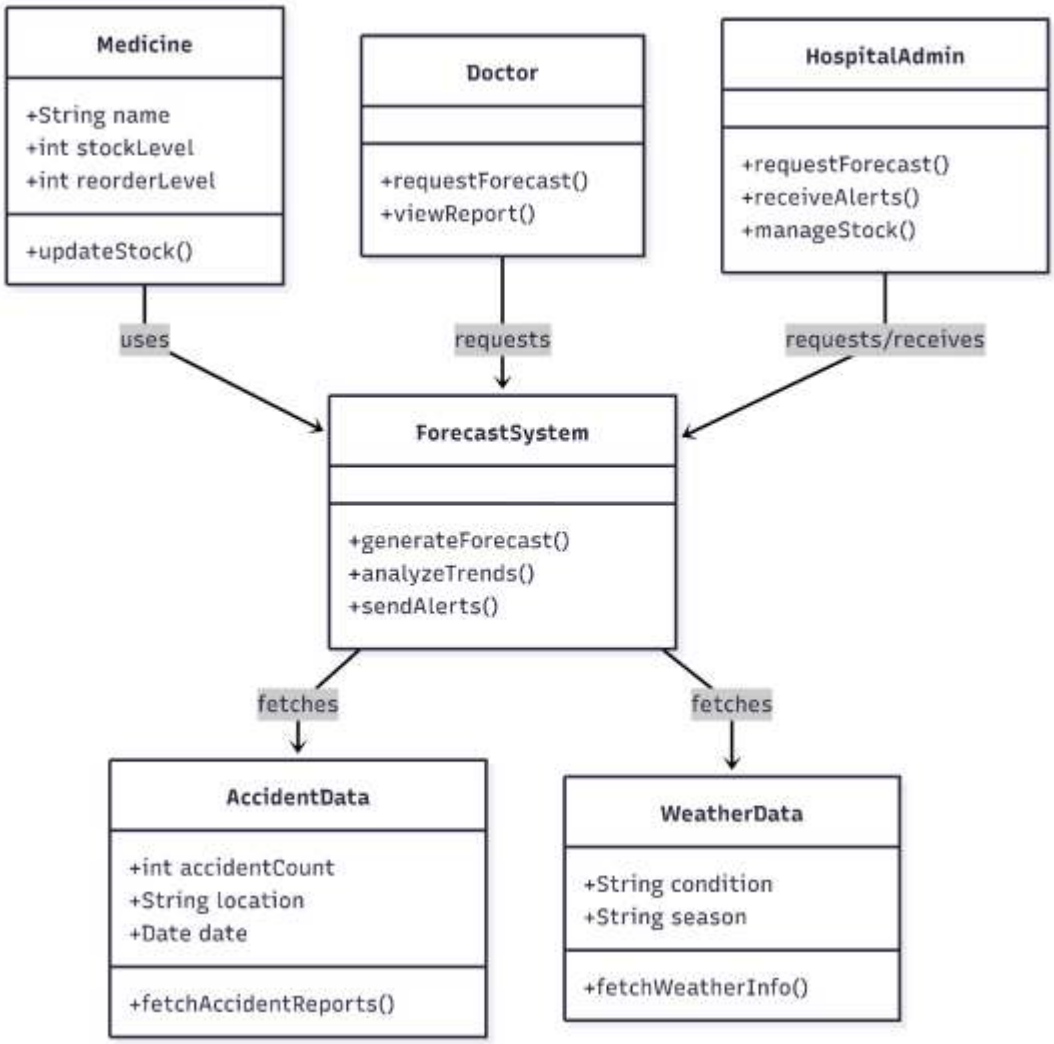

Figure 5 : Class Diagram

System Workflow

A sequential process dictates the following workflow of this system:

- Hospital records are used to gather data
- Data is preprocessed and cleaned
- Machine learning models are trained
- Predictions are generated
- Surge detection is performed
- Results are adjusted dynamically
- User see outputs

2.3.3. Model Development and Integration

At the heart of the system, we have built our core functionality based on combining two machine learning models: Random Forest and XGBoost.

2.3.3.1. Random Forest Demand Forecasting Model

Predicting Medicine Demand in Normal Conditions using REST API on Top of Random Forest Algorithm Random Forest Classifier is a type of ensemble learning that creates multiple decision trees and aggregates the predictions to boost accuracy and limit overfit. Historical data on medicine usage were used to train the model, enabling it to learn, among others:

- Daily and weekly frequency of use
- Seasonal variations
- Consumption fluctuations

Baseline Demand Forecasting: The trained model produces the demand predictions, which form the basis of your forecasting system.

2.3.3.2. XGBoost Model for Accident Surge Detection

We implemented an XGBoost model to help identify potential spikes in admissions related to accidents. Using accident data, this model attempts to determine patterns responsible for sudden surge of cases requiring emergency treatment. The model is capable of:

- Identifying high-risk periods
- Detecting abnormal spikes in accident cases
- Writing conditions in terms of normal/surge

Surge condition is defined as the predicted number of accident cases exceeds a pre-specified limit based on historic trends.

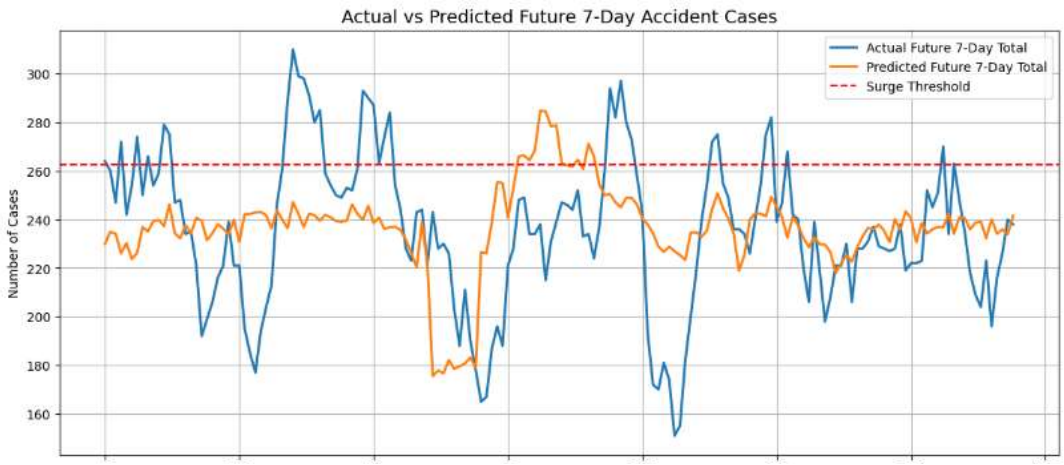

Figure 6 : Actual vs Predicted Accident Cases

2.3.3.3. Model Integration Strategy

A key innovation of the proposed system is the integration of the Random Forest and XGBoost models. Initially, the Random Forest model generates a baseline prediction for medicine demand. The XGBoost model then evaluates accident data to determine whether a surge condition exists. If a surge is detected, the system dynamically adjusts the predicted medicine demand to reflect increased requirements. Otherwise, the baseline prediction is used. This integration ensures that the system can handle both routine and emergency scenarios effectively.

2.3.4. Implementation

Implementation phase–during this phase, all the modules of system integrated into a single working application. The backend sends requests to the machine learning models for prediction, whereas the frontend presents the result in an interactive way. The system allows users to:

- View predicted demand values
- Monitor trends through visualizations
- For surge conditions, alert alerts
- Dashboard interface
- Prediction results page
- Alert notifications

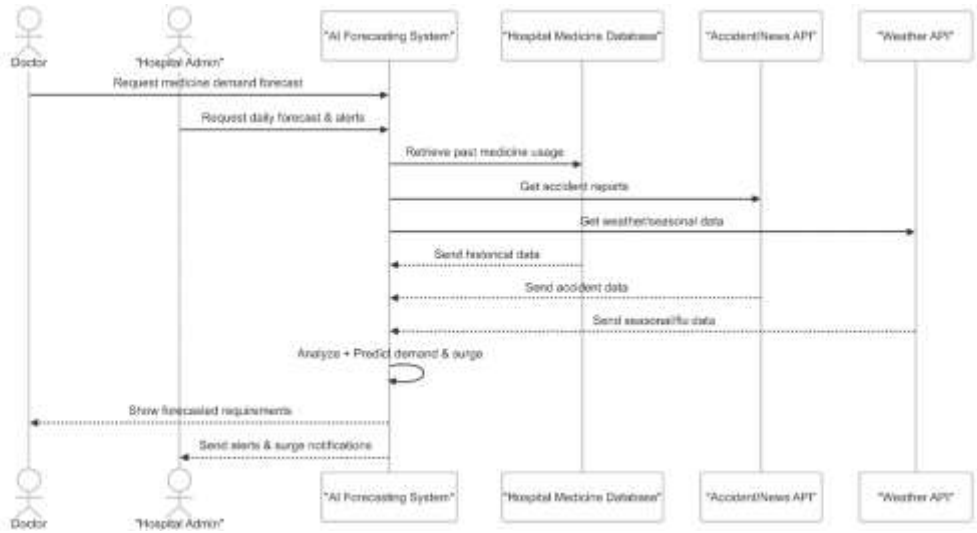

Figure 7 : Sequence Diagram

Key System Features

System components: The devised system has a number of features which are as following:

- Machine learning–based demand forecasting
- Accident surge detection
- Dynamic adjustment of predictions
- Real-time data visualization
- User-friendly interface

All of this functionality combined enables better decision making and optimizes hospital resource utilization.

Summary

The application development successfully combines machine learning techniques and web-based system to offer an intelligent solution for hospital resource optimization. Thus, by merging predictive analytics and the practical implementation of this technology, it ensures users with accuracy alongside usability in real-time scenarios on ground.

2.4. Testing

Since system testing is a vital part of the entire proposed medicine demand forecasting and accident surge detection system development life cycle, designed to play an important role in helping ensure that the solution performs reliably, accurately, and efficiently in real-world healthcare practice. In this research, testing is not just for the web-based application developed but also to check if integration of any machine learning model gets meaningful and actionable predictions.

The proposed solution implements a set of modules/features that are not simply individual software systems — for example, data preprocessing modules, machine learning models, backend services, database operations and frontend visualization layers. Hence, the way the testing was conducted had been on a whole to test the functionality of each component in isolation and also by integrating all components together with that of the system.

The reason is that the system needs to deal with various data such as historical records of medicine usage and admission data related to accidents, but it can also produce irrelevant predictions when circumstances change. The testing was conducted on real-world datasets derived from hospital records that tested system performance in realistic operational conditions. This ensures that the results obtained based on research can be generalized over real clinical settings rather than confined to theoretical validation.

In this research, testing has a large part in evaluating the machine learning models inside the system. The Random Forest model which predicts medicine usage has to produce training signals that are in close alignment with actual patterns of usage. Concerns with the XGBoost Model Likewise, in accident surge detection using XGBoost model it is paramount to correctly identify an abnormal rise in emergency cases. The validation of these models is critical, as any inadequacies in these models might translate directly to decision making in the hospital setting.

Besides the accuracy of the model, the system was tested for dynamic prediction adjustment based on surge conditions detected. In fact, one of the main contributions of the research is this integration since it makes your environment respond in a routine as well as emergency situations. As such, testing was set up to ensure this adjustment mechanism is operating properly in different circumstances.

System Performance was tested, assessing parameters such as data processing time, response time and stability (especially when handling larger datasets). The system is used for hospital environments; therefore, it needs to work consistently without lags and failures. And this helps to make sure that how this system will handle the real-time or near real time data processing requirement.

In addition, further usability testing was done to ensure a good user experience. Healthcare administrators who are not technically minded should be able to read results of prediction and alerts easily. As such, we deemed the clarity of visualizations, readability of outputs and the general user experience to be key testing criteria. For a thorough assessment, multi-level testing was the method of choice. Such as unit testing, integration testing, system testing and user interface testing. They target different layers of the system and guarantee proper functionality of each component both in isolation and as part of a complete set.

In general, the test phase is a critical one to validate as well verify that the new system meets their assigned functional and non-functional requirements defined above. Testing verifies that given an appropriate software component and machine learning model, the entire system is able to support accurate, reliable, and efficient hospital resource management.

2.4.1. **Test Plan**

Test Plan RawData: Test plan defines the approach, objective, scope and estimation methodology of MDMFA that is proposed medicine demand forecasting and accident surge detection system. It acts as a template for performing testing in an organized and controlled manner, securing that everything within the system is validated.

Since the system being built brings together software components and machine learning models, we created a multi-faceted test plan including needs for data validation, checking model accuracy, proper functioning of the system, expected integration behavior and user interaction. The aim is to make sure that under both normal conditions and emergency cases the system functions reliably, providing correct predictions and outputs of serious significance for health care decision making.

2.4.1.1. **Testing Objectives**

The most important goals of the testing process are defined as follows:

- To assure the system has been programmed to process and save input datasets correctly that contain medicine usage data along with Accident admission records.
- To examine Random Forest model to predict the demand of medicines.
- To evaluate the prediction of surge condition caused by accidents using XGBoost model.
- To verify that the system is properly combining both models in a way that predictions change dynamically depending on whether surge conditions are detected.
- To check whether backend APIs perform and return responses the right way for data requests.
- Safeguarding frontend interface outputs ranging from clear to accurate.
- For measuring the performance of a system with respect to response time, scalability and stability.
- To make sure that the system satisfies functional and non-functional requirements defined in earlier phase.

These goals help to ensure that the system is technically accurate as well as useful in hospital administrator applications.

2.4.1.2. Testing Scope

Testing Scope is used to define the limits of the testing and what components of systems are included in the evaluation. Testing was performed within the following scope:

- Data input module - Dataset uploaded by user (validation, will be saved).
- Data Preprocessing Module - Validation of data cleansing, transformation and feature extraction
- Machine Learning Models - Testing predictions and surge-detection models
- Integrated Models - Testing Random Forest to XGBoost models
- Backend System - API Functionality and Server-side Processing Verification
- Database Operations - Of writing, reading, and MongoDB consistency
- Frontend Interface - Assessment of dashboard visualization, charts, and alerts

Test not scoped– Following were considered as out of scope to be tested

- External integrations with various hospital systems (if any new integrations to be added)
- Testing in a live hospital environment for long term deployment

This limited scope guarantees testing of all essential parts that comprise the proposed system without leading to overburden on our testing process.

2.4.1.3. Testing Approach

A focused testing strategy was put in place to guarantee a thorough review of the system. The approach involves different layers of testing, where each layer targets a specific component of the overall functionality of the system.

- Black-box Testing - This testing is done to evaluate system outputs depending on given inputs without analyzing the internal code structure. These methods were primarily used for prediction output and system behavior validation only.
- White-box Testing - This is done at the development level and tests internal logic, especially for data preprocessing or backend operations.
- Model Validation Test - Used for the direct testing of machine learning models measured by statistical performance metrics.
- Scenario Based Testing - Simulation of real-world conditions (for example, accident surge conditions) was done to check how the system responds.

This combination of testing strategies assures both logical correctness and predictive accuracy.

2.4.1.4. Testing Environment

The testing environment identifies the hardware and software configuration on which the testing activities are performed.

- Hardware Environment:
  - Personal Computer with at least 8GB RAM
  - Regular CPU Used for Executing machine Learning Models
- Software Environment:
  - OS: Windows 10 / Windows 11 (64-bit only)
  - Backend: Node. js with Express
  - Frontend: React.js
  - Database: MongoDB
  - Machine Learning: Python (Scikit-learn, XGBoost)
  - Browser are you using: Google Chrome / Microsoft Edge

This environment provides a testing ground that resembles real-world deployment environments.

2.4.1.5. Test Data

Testing was conducted using real-world datasets collected from Yaggahapitiya Divisional Hospital. These datasets include:

- Medicine usage dataset (approximately 16,500 records)
- Accident admission dataset (approximately 10,000 records)

Using real data ensures that the system is tested under realistic conditions, improving the reliability of the results.

2.4.1.6. Expected Outcomes

The expected outcomes of the testing process are as follows:

- The system should successfully process and store input datasets without errors.
- The machine learning models should generate accurate and consistent predictions.
- The system should correctly detect accident surge conditions.
- The prediction adjustment mechanism should function correctly under surge scenarios.
- The system should display outputs clearly through the user interface.
- The system should maintain stable performance under different workloads.

Achieving these outcomes confirms that the system meets its intended objectives and is suitable for real-world application.

2.4.1.7. Risks and Assumptions

The testing process is subject to certain risks and assumptions:

- Risk: Incomplete or inconsistent data may affect model accuracy
- Risk: System performance may vary under extremely large datasets
- Assumption: Input datasets are representative of real hospital scenarios
- Assumption: Users have basic knowledge of interpreting system outputs

Identifying these factors helps in understanding potential limitations and ensuring better system evaluation.

2.4.1.8. Summary of Test Plan

The test plan provides a comprehensive framework for evaluating the proposed system. By defining clear objectives, scope, testing approaches, and expected outcomes, it ensures that the system is tested systematically and thoroughly. This structured approach enhances the reliability of the testing process and supports the validation of both system functionality and machine learning performance.

2.4.2. Test Strategy

The test strategy defines the detailed approach used to execute testing activities for the proposed medicine demand forecasting and accident surge detection system. While the test plan outlines what needs to be tested, the test strategy explains how the testing is carried out in a systematic and effective manner.
Given the hybrid nature of the system, which combines machine learning models with a web-based application, the testing strategy was designed to address both software functionality and predictive model performance. The strategy ensures that each component is tested thoroughly before integration and that the complete system operates reliably under different conditions.

2.4.2.1. Testing Approach Overview

An incremental, layered testing approach was taken for this research. This way, we can test each component in isolation before bringing them together into the entire system. They ensure that the chances of failures happening across the system are lower by identifying errors in the early stages. We performed testing in the following different important levels:

- Unit Testing
- Integration Testing
- System Testing
- Model Testing
- Performance Testing
- User Interface Testing

By validating at each level of the system, these tests ensure full coverage.

2.4.2.2. Unit Testing

We used unit testing, to test that each component of the system worked as applicably expected. Each module was tested independently to verify it behaved as designed. During unit tests, the following components were checked:

- Functions for data preprocessing (cleaning Missing Values, Duplicates etc.)
- Feature Extraction Processes (date-based features like day and month)
- Backend API endpoints
- Database operations (insert, update, retrieve)

Unit testing was designed to verify that each surface works correctly for a given equivalent input. This is important to prevent errors from propagating to higher tiers of the system.

2.4.2.3. Integration Testing

After unit testing was done, integration testing was performed to see how the system parts worked together. As the proposed system includes several interconnected modules, it is necessary to provide them with seamless communication. The following integrations were tested:

- API for communication between frontend backend
- Backend - Database interaction
- Communication between Backend and Machine Learning model
- Random Forest and XGBoost models integrated
- Integration testing - where the aim was to verify data flow throughout the system and that outputs from one module were correctly consumed as input by another.

2.4.2.4. System Testing

We performed system testing in order to test the full system as a complete unit. This stage checks if the system is fulfilling all functional requirements and hence working as expected in actual conditions. The system was subjected to end-to-end workflow simulation which included:

- Uploading datasets
- Running preprocessing
- Generating predictions
- Detecting surge conditions
- Adjusting medicine demand
- Displaying results on the dashboard

System testing is performed to check its working with all components flawlessly and the outputs it generates.

2.4.2.5. Model Testing

Here model testing specifically aims at performance of machine learning models which are used in the system

- Random Forest Model Testing

Historical medicine usage data were used to evaluate the Random Forest model. Regression metrics MAE, RMSE, and WAPE are used to evaluate the accuracy by comparing actual values with predictions.

- XGBoost Model Testing

Accident admission data were used to assess the potential of the XGBoost model for detecting surge conditions. Test data were per metric (accuracy, precision, recall and F1 score. Notably, recall was emphasized because failure to detect a surge condition in healthcare could have dire ramifications.

2.4.2.6. Performance Testing

Performance tests were performed to assess the capacity of the system when dealing with large datasets and multiple user requests. We tested the system under several different conditions, including:

- Large dataset processing
- Multiple API requests
- Continuous prediction execution

The goal was to make sure the system runs constant under no delays or failures.

2.4.2.7. User Interface Testing

User interface testing was done to make the system non-complex in usage, and output clear. The following aspects were evaluated:

- Dashboard layout and navigation
- Clarity of prediction results
- Visualization of graphs and charts
- Display of alerts and notifications

This system is aimed at healthcare administrators; therefore, usability and readability were of prime importance.

2.4.2.8. Error Handling and Edge Case Testing

Similar evaluations of the system were also performed across various edge-cases and for unexpected inputs. Examples include:

- Missing or incomplete datasets
- Invalid data formats
- Extremely high or low values
- No accident data scenarios

The whole point of the system was that it would deal with such cases cautiously giving out error messages or outputs.

2.4.2.9. Automation vs Manual Testing

It employs a mix of manual testing and automated testing techniques. Used for validating user interfaces and scenarios Manual Testing. Automated Testing for validation of backend API and model evaluation repetitively. This combination provides ample flexibility, and also ability to run tests at a faster pace.

2.4.2.10. Summary of Test Strategy

The mixed testing levels and approaches make sure all aspects of the proposed system are evaluated comprehensively in a test strategy. This approach meets the goals of validating software functionality and machine learning performance, and this creates a strong foundation and can work like a blueprint for how to validate the system overall.
This formalized approach helps ensure the system can run predictably under real-life conditions and facilitate decision making based on data regarding hospital resources.

2.4.3. Test Cases

Test cases were designed to validate the functional behavior, data processing capabilities, machine learning performance, and overall system integration of the proposed medicine demand forecasting and accident surge detection system. Each test case represents a specific scenario that the system is expected to handle, ensuring that all core functionalities are thoroughly verified.

The test cases were developed based on the system requirements and real-world hospital scenarios. These include both normal operating conditions and emergency situations, such as sudden accident surges. The testing process also considered edge cases, invalid inputs, and performance-related conditions to ensure system robustness.

Table 2 :Random Forest Prediction Accuracy

Test Case ID	**TC_01**
Test Case Name	Random Forest Medicine Demand Prediction
Objective	To verify whether the Random Forest model predicts the next-day medicine demand accurately.
Input	Historical medicine usage data
Steps	Run the Random Forest prediction model using preprocessed medicine usage data.
Expected Result	The system should generate next-day predicted medicine quantities for selected medicines.
Actual Result	The system generated next-day medicine demand predictions successfully.
Status	Pass

Table 3 :XGBoost Surge Detection

Test Case ID	**TC_02**
Test Case Name	Accident Surge Detection
Objective	To verify whether the XGBoost model detects accident surge conditions correctly.
Input	Accident admission data
Steps	Run the XGBoost model using accident-related admission records.
Expected Result	The system should classify the condition as either normal or surge.
Actual Result	The system detected the accident condition successfully.
Status	Pass

Table 4 : Demand Adjustment During Surge

Test Case ID	**TC_03**
Test Case Name	Dynamic Medicine Demand Adjustment
Objective	To verify whether medicine demand is adjusted when an accident surge is detected.
Input	Random Forest prediction output and XGBoost surge detection output
Steps	Run the integrated forecasting module under a surge condition.
Expected Result	The system should increase the predicted medicine demand according to the detected surge condition.
Actual Result	The system adjusted the predicted medicine demand successfully.
Status	Pass

Table 5 : Normal Condition Prediction

Test Case ID	**TC_04**
Test Case Name	Baseline Medicine Demand Prediction
Objective	To verify whether the system uses the normal Random Forest prediction when no surge is detected.
Input	Random Forest prediction output and normal accident condition
Steps	Run the integrated forecasting module under a non-surge condition.
Expected Result	The system should display the baseline next-day medicine demand prediction without adjustment.
Actual Result	The baseline medicine demand prediction was displayed successfully.
Status	Pass

Table 6 : Daily Stock Reduction After Medicine Usage

Test Case ID	**TC_05**
Test Case Name	Calculate Remaining Stock After Daily Usage
Objective	To verify whether the system calculates the remaining stock after deducting the daily medicine usage from the current stock.
Input	Current stock count and next-day predicted medicine usage count
Steps	Retrieve the predicted medicine count and subtract it from the available stock quantity.
Expected Result	The system should calculate and display the correct remaining stock quantity for each medicine.
Actual Result	The remaining stock was calculated and displayed successfully.
Status	Pass

Table 7 :Daily Automated Stock Monitoring

Test Case ID	**TC_06**
Test Case Name	Daily Stock Monitoring Process
Objective	To verify whether the system performs daily stock calculation using predicted medicine demand.
Input	Daily predicted medicine demand and current stock records
Steps	Execute the daily stock update process.
Expected Result	The system should update remaining stock values daily after applying predicted medicine usage.
Actual Result	The system updated the daily remaining stock values successfully.
Status	Pass

Table 8 : Threshold-Based Stock Order Generation

Test Case ID	**TC_07**
Test Case Name	Automatic Stock Order Creation
Objective	To verify whether the system creates a new stock order when the remaining stock falls below the threshold value.
Input	Remaining stock quantity and predefined threshold value
Steps	Run the stock monitoring process after daily usage calculation.
Expected Result	If the remaining stock is below the threshold value, the system should automatically generate a stock order request.
Actual Result	The system generated a stock order request successfully when stock went below the threshold.
Status	Pass

Table 9 :No Stock Order Above Threshold

Test Case ID	**TC_08**
Test Case Name	Prevent Unnecessary Stock Order
Objective	To verify whether the system avoids creating a stock order when remaining stock is above the threshold value.
Input	Remaining stock quantity greater than threshold value
Steps	Run the daily stock monitoring process.
Expected Result	The system should not generate a new stock order when stock is sufficient.
Actual Result	No stock order was generated when stock was above the threshold.
Status	Pass

Table 10 :Pharmacist Stock Update

Test Case ID	**TC_09**
Test Case Name	Update Medicine Stock by Pharmacist
Objective	To verify whether the pharmacist can update medicine stock after receiving new stock.
Input	Medicine name and updated stock quantity
Steps	Pharmacist enters the received stock quantity and updates the medicine inventory.
Expected Result	The system should update the stock quantity correctly in the database and display the latest available stock.
Actual Result	The pharmacist stock update was completed successfully.
Status	Pass

Table 11 : Database Storage

Test Case ID	**TC_10**
Test Case Name	Store Prediction and Stock Records
Objective	To verify whether prediction results, remaining stock values, and stock order records are stored correctly.
Input	Prediction output, stock calculation result, and stock order request
Steps	Run prediction and stock monitoring process.
Expected Result	The system should save all relevant records in the database.
Actual Result	Records were stored successfully.
Status	Pass

Table 12 : Database Retrieval

Test Case ID	**TC_11**
Test Case Name	Retrieve Medicine and Stock Records
Objective	To verify whether stored medicine records, prediction results, and stock order details can be retrieved correctly.
Input	Database retrieval request
Steps	Request stored records from the database.
Expected Result	The system should retrieve accurate records without data loss.
Actual Result	Records were retrieved successfully.
Status	Pass

Table 13 : API Functionality

Test Case ID	**TC_12**
Test Case Name	Backend API Response Validation
Objective	To verify whether backend APIs return correct responses for prediction, stock, and order functions.
Input	API request from frontend
Steps	Send API requests for prediction, stock update, and stock order retrieval.
Expected Result	The backend should return valid responses with correct status codes and data.
Actual Result	APIs returned correct responses successfully.
Status	Pass

Table 14 : Dashboard Display

Test Case ID	**TC_13**
Test Case Name	Dashboard Data Display
Objective	To verify whether prediction, stock, and order details are displayed correctly on the dashboard.
Input	Prediction values, stock values, and order details
Steps	Open the dashboard after running the prediction and stock monitoring process.
Expected Result	The dashboard should clearly display predicted demand, remaining stock, threshold status, and order details.
Actual Result	Dashboard displayed all details successfully.
Status	Pass

Table 15 : Graph Visualization

Test Case ID	**TC_14**
Test Case Name	Prediction and Stock Graph Display
Objective	To verify whether the system displays medicine prediction and stock trend visualizations correctly.
Input	Historical usage, predicted demand, and stock level data
Steps	Open the visualization section of the dashboard.
Expected Result	The system should display readable graphs showing medicine demand and stock trends.
Actual Result	Graphs were displayed successfully.
Status	Pass

Table 16 : Alert Notification

Test Case ID	**TC_15**
Test Case Name	Low Stock and Surge Alert Display
Objective	To verify whether the system displays alerts for accident surge conditions and low stock levels.
Input	Surge detection output and remaining stock below threshold
Steps	Run the system under surge condition and low-stock condition.
Expected Result	The system should display appropriate warning alerts for surge and low-stock situations.
Actual Result	Alert notifications were displayed successfully.
Status	Pass

Table 17 :Concurrent Users

Test Case ID	**TC_16**
Test Case Name	Concurrent User Access
Objective	To verify whether the system remains stable when multiple users access prediction and stock details at the same time.
Input	Multiple simultaneous dashboard/API requests
Steps	Access the system from multiple sessions and request prediction, stock, and order details.
Expected Result	The system should respond correctly without crashing or displaying incorrect data.
Actual Result	The system handled concurrent access successfully.
Status	Pass

2.4.3.1 Summary of Test Cases

The test cases listed above validate the core workflow of the proposed medicine demand forecasting and accident surge detection system. The testing process confirms that the system can predict next-day medicine demand, detect accident surges, adjust medicine requirements, calculate remaining stock, generate stock orders when stock levels fall below the threshold, and allow pharmacists to update medicine inventory after receiving new stock. In addition, database operations, API communication, dashboard display, graph visualization, alert notifications, and concurrent user access were tested to ensure that the system operates reliably as a complete application.

3. RESULTS & DISCUSSION

The results of the proposed medicine demand forecasting and accident surge detection system are presented in this chapter. This chapter serves to assess if the machine learning models developed, and the working system integrated, actually gave away from as stated objectives during research. Results are presented with respect to medicine demand forecasting accuracy, accident surge detection performance, dynamic adjustment of demand and stock monitoring, as well as assessing the overall usefulness of the system.

Two structured datasets gathered from hospital records were used to test the proposed system. The first dataset was about daily medicine usage, and the second one was related to admission records due to accidents. Future medicine demand forecasting (using Random Forest model, accident burst condition detection using XGBoost model) The results we get using these models show how machine learning can aid. to active hospital resource planning.

3.1. Medicine Demand Forecasting Results

We built the component of the medicine demand forecasting using a Random Forest regression model. This model was mainly to predict next day medicine demand from historical usage pattern of that medicine. Daily medicine usage data from hospital pharmacy and ETU records were used to train the model.

The forecasting medicines selected were high use ETU medicines. These medicines were chosen as they are some of the most relevant to emergency medicine and have a direct effect on patient treatment. The model performance was evaluated using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Weighted absolute percentage error (WAPE) and Accuracy.

Table 18 :Performance Comparison for Medicine Demand Forecasting

Medicine	**MAE**	**RMSE**	**WAPE (%)**
Inj. Hydrocortisone 100mg	3.22	4.82	8.95
Inj. Omeprazole	0.87	2.24	3.23
Inj. Atropine	0.27	1.01	2.12
Inj. Adrenaline 1:1000	1.17	2.22	8.52
Overall Model	**0.64**	**1.84**	**4.55**

The overall model achieved a WAPE value of **4.55%**, indicating that the forecasting error was relatively low. Based on WAPE, the overall forecasting accuracy can be calculated as:

- Accuracy = 100 − WAPE
- Therefore:
- Accuracy = 100 − 4.55 = 95.45%

This result shows the Random Forest performed well as a model applied to predict medicine demand. The low total MAE and RMSE values indicate that predicted quantities were approximate to actual medicine usage values.

Among the selected medicines, Inj. The performance of the models was evaluated with MAE and WAPE metrics, where atropine achieved the lowest values in both cases, which contributes to its prediction stability [20]. This could be due to the pattern of Inj. usage Atropine was found to be more uniform than other medications. On the other hand, Inj. Hydrocortisone 100mg and Inj. This resulted in a relatively higher WAPE when such cases were included during analysis for Adrenaline 1:1000. This is the result of an irregular use of emergency departments, in which demand can fluctuate considerably, depending on illness or trauma due to accidents.

In conclusion, these results indicate that the Random Forest model is appropriate for forecasting medicine demand in a hospital setting. By estimating future needs of medicine requirements, forecasting accurately helps hospital staff predict prescription volume at different times to avoid unnecessarily overstocking and emergency shortages.

3.2. Accident Surge Detection Results

Accident surge detection is implemented using XGBoost model. This model was designed to detect spikes in accident admissions that might affect demand for emergency medicine. The model was trained on data of admission on accident, along with information about type of accidents, number cases answered for each type of accident case admitted and how many types of injuries were involved in the accident at a time and medicines consumed.

I would suggest to use both regression and classification related metrics to evaluate the model. Such metrics were employed to evaluate how well the model identified scenarios indicating an imminent accident surge.

Table 19 :Performance Comparison for Accident Surge Detection

Metric	Value
MAE	254.30
RMSE	314.14
Accuracy	0.81
Precision	0.82
Recall	0.99
F1-score	0.90

The accuracy for the XGBoost model is 0.81, which indicates that the model was able to classify a large number of normal and surge conditions correctly. The precision score of 0.82 indicates that for most of the egas cases predicted to be surge conditions were indeed relevant surge cases

This recall value of 0.99 is very significant in this study. If an accident spike is not detected at a hospital, the results can be dire -- woe to those who cannot obtain necessary medicine in time, and welcome death from delayed treatment. So, a very high Recall value is very useful because it means that the model was able to detect as many actual surge situations.

With the F1-score of 0.90, we observe a healthy harmony between precision and recall. This validates that the model is robust when it comes to estimating emergency demand patterns due to accidents.

Even though, the MAE and RMSE values are on the high scale, these can be justified as its accident data. Admission numbers of accidents can fluctuate a lot depending on the unpredictable outside factors (weather, public holidays, weekends etc., and any possible emergency events). Hence, some portion of prediction variance is expected.

3.3. Dynamic Medicine Demand Adjustment Results

This research contributes to the solid body of work focused on medicine demands forecasting and accident surge requests detection. The baseline medicine demand prediction is provided by the Random Forest model and whether an accident surge condition occurs is identified using the XGBoost model.

If a surge is not detected, the baseline medicine demand prediction is then used by the system. However, when a spike in demand is identified, the system updates its prediction of medicine needs to account for the additional requirement created by emergency admissions.

The integrated method is more realistic than utilizing medicine demand forecasting alone, given the highly impact of road accident patient inflow upon emergency medicine usage. Thus, for example, if the Random Forest model predicts a normal demand value for a medicine but the XGBoost model signals an accident surge, the system corrects up the expected demand to trigger reinforce their emergency stats monopoly.

Table 20 : Demand Adjustment Logic

Condition	System Action
No accident surge detected	Use baseline Random Forest prediction
Accident surge detected	Increase predicted medicine demand
Remaining stock above threshold	No stock order generated
Remaining stock below threshold	Automatic stock order generated

The results show that the integrated system can support both normal and emergency scenarios. This makes the system more adaptive compared to traditional forecasting approaches that only depend on historical average usage.

3.4. Stock Monitoring and Reorder Results

The system does not limit to just forecasting and surge detection; It also calculates the after-use remaining stock of medicines for every day. From the predicted next-day medicine count, which is a prediction of future expected usage, this predicted quantity is deducted from the current needed amount of that particular medicine. This left is calculated as follows:

Surplus Stock = Existing Out of Supply stock − Anticipated Drugs Use

When remaining stock drops below the specified threshold, an automated stock order request is made by the system. This feature, in turn, encourages preventive care management and lowers the likelihood of stockouts or loss of drugs.

This process ensures that medicine inventory is continuously monitored. It also reduces manual workload for hospital pharmacy staff because the system automatically identifies low-stock situations.

The pharmacist update function further improves the system workflow. After new medicine stock is received, the pharmacist can update the system with the latest quantity. This ensures that future predictions and stock calculations are based on updated inventory records.

3.5. System Output Discussion

The outputs that the system provide are medicine demand forecast, accident surge level, orders placed and threshold alerts. These outputs are shown on the browser-based dashboard to help hospitals in their decision-making process. The dashboard helps users understand:

- Expected next-day medicine demand
- Whether an accident surge condition exists
- Current remaining stock level
- Whether stock is below threshold
- Whether a stock order has been generated

Visually, graphs are a major improvement in interpretability. Members of the hospital staff don't have to go examine raw data manually, but rather can parse trends and alerts through the interface. This enhances ease-of-use and helps in quicker decision-making.

3.6. Discussion

This research shows that the proposed system can be used to improve medicine demand forecasting and emergency resource planning. The overall accuracy achieved by the Random Forest model is 95.45%, which shows that the model can predict medicine usage based on historical medicine consumption patterns quite well.

The XGBoost model generally detected the surges in accidents with a high recall value at 0.99, and an F1-score of 0.90. High recall is crucial because the system aims to prepare for emergencies. Here, the priority is on identifying 90 percent of surge occurrences versus missing a few high-profile emergencies.

This is a significant benefit of integrating the two models. In emergencies, medicine demand forecasting might not be enough because the number of patients admitted due to accidents can spike suddenly. Using forecasting along with detecting a surge, the system will become smarter and more realistic.

That is just more of the functional utility of the system with stock tracking and automatic order. Not only does it predict demand, but the system also assists with inventory decisions by effectively assessing inventory and creating reorder requests for stocks that fall short of levels. The advantages of the proposed system over conventional manual estimation methods are listed as follows:

- More accurate medicine demand prediction
- Better emergency preparedness
- Reduced risk of medicine stockouts
- Reduced risk of carrying excess inventory and spoilage
- Faster decision-making through dashboard visualization
- Enhanced support of pharmacy administrators and hospital

But the system does to have its pitfalls. A major factor affecting the model performance is quality and completeness of historical data Inspiring Incidents: If data has missing values, erroneous entries or anomalous patterns prediction accuracy can go wrong. Finally, the current system is built on a limited choice of ETU medicines and accident type categories meaning that the next step is to apply the system to further medicines and departments as well as external drivers such as seasonal effects, public holidays and traffic patterns.

3.7. Summary

Results and discussions of the proposed medicine demand forecasting and accident surge detection system were presented in this chapter. The Random Forest model performed well in predicting medicine demand, and the XGBoost model performed well, in detecting accident surge conditions.

The interfaced system updated the predicted medicine demand according to surge conditions and monitored stock levels by calculating remaining stocks, checking these against lower and upper threshold limits, generating automated stock orders based on all available predictive cues, and updating pharmacist stock entries.

In general, results confirm that the proposed system can optimize hospital resource management in terms of: i) supporting proactive planning of medicine orders on a monthly basis; ii) Minimize the chances of experiencing critical shortages or unavailability of proper medicines and iii) Enhance emergency preparedness.

4. CONCLUSION

This research is based on the framework for medicine demand forecasting with machine learning system and how to detect a surge of accidents, which could help in the management of hospital resources. The research resolved an inconvenient concern for healthcare organizations, which was how to approximate the proper equilibrium confined between supply of medicine and patient demand in uncertainty emergency situations.

The designed system combines two machine learning models for demand prediction of routine medicine (Random Forest model) and accident surge detection (XGBoost model). By combining these two models, the system can adapt medicine demand forecasting in real time by utilizing available conditions, resulting in a forecasting method that offers more accuracy and adaptability compared to traditional methods.

The outcome of the study is that Random Forest model predict well and the overall accuracy is around 95.45%. This implies that the model could are capable of capturing past consumption trends and make accurate predict for future requirement of medicines. Likewise, detection of the accident surge condition from the XGBoost model showed robustness as a result of a high recall value which is an important metric for recognizing an emergency situation in a health care environment.

The system also has besides predictive capabilities it consists of Stock monitoring mechanism which computes remaining medicines stock through analysis of predicted utilization. If the stock level goes below a certain limit, the system raises a request for stock order on its own. This strengthens usage-based inventory management and helps prevent drug shortages. Additionally, the pharmacist-mediated stock updates ensure that only accurate data is remaining at all times anytime new stock comes in.

Having forecasting, surge detection and stock management together gives a holistic solution to hospital resource optimization. Considering classical manual methods that are based on static estimates, the implemented system in this proposal can provide data-driven and intelligent decision-making support by driving users to act proactively. It improves usability further as a web-based implementation for Health care administrators to get the predicts, monitor trends and respond to alerts in an efficient way.

Although the results look favourable, the study does have some limitations. System works well as long as Historical data are Deep either in Depth or Quality Dispersion of the dataset also plays an essential role in how accurately can it predict from the new data. It currently uses a narrow range of medicines and accident types. One can imagine future developments where more data sources are added, such as weather

conditions, public holidays, and rush hour traffic patterns to help refine predictions even further.

Finally, this proposed system provides an excellent capability of using machine learning for changing the management of hospital resources. The study demonstrates a scalable and practical forward-looking approach to utilize predictive analytics with system implementation that promises not just greater availability of medicines but also proactive and reactive emergency preparedness with efficient health-care operations. This system can also be extended to more general hospital settings and is an important addition to health informatics.

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6. APPENDICES